

AI-Based Smart Attendance & Academic Management System

Face Recognition + QR Code + Manual Attendance

1. Project Objective

The objective of this project is to design and implement a scalable, secure, and intelligent attendance and academic management system that can automatically mark attendance using face recognition, provide secure QR-code based fallback attendance, and allow manual attendance as a last option. The system supports Admin, Teacher, and Student roles with role-based access control.

2. System Overview

The system is divided into three main modules: Admin Web Application, Teacher Web Application, and Student Mobile Application. Attendance can be marked using three modes: Face Recognition (primary), QR Code Scan (secondary), and Manual Attendance (tertiary).

3. Admin Module (Web Application)

- Register and manage students with unique register numbers and face data
- Register teachers and assign subjects and sections
- Create departments, semesters, sections, and subjects
- Create and manage section-wise timetables with teacher availability checks
- View, verify, and lock attendance records
- View, verify, and publish marks and results
- Send notifications to students and teachers
- Configure attendance rules and role-based access control
- Maintain audit logs and system backups

4. Teacher Module (Web Application)

- View assigned sections, subjects, and student lists
- View personal and section-wise timetable
- Take attendance using face recognition in classroom
- Generate dynamic QR codes for attendance fallback
- Mark attendance manually when required with reason
- Assign internal, assignment, and lab marks
- Send notifications to assigned sections
- Use offline attendance mode and sync later

5. Student Module (Mobile Application)

- View overall and subject-wise attendance percentage
- View real-time attendance status during class
- Scan QR code and submit selfie for attendance
- View marks and final results

- View daily and weekly timetable
- Receive notifications from admin and teachers
- View personal academic profile

6. Attendance Modes

Face Recognition Attendance uses a classroom camera to detect and recognize multiple student faces simultaneously. QR Code Attendance uses a time-bound dynamic QR code with selfie verification. Manual Attendance allows teachers to mark attendance manually with justification.

7. Security and Anti-Proxy Measures

- Face recognition confidence thresholding
- QR code expiry and one-time usage per session
- Selfie verification for QR attendance
- One attendance record per student per class
- Role-based access control
- Encrypted storage of face embeddings
- Complete attendance audit trail

8. Database Design (Core Tables)

- Users (Admin, Teacher, Student)
- Students
- Teachers
- Departments
- Sections
- Subjects
- Timetable
- Attendance
- Marks
- Notifications
- Audit Logs

9. Technology Stack

Backend: FastAPI or Node.js, Face Recognition using InsightFace/OpenCV, PostgreSQL database, Redis cache. Frontend: React for Admin and Teacher web applications, Flutter for Student mobile application. Infrastructure: GPU-enabled server, Docker-based deployment.

10. Complete System Workflow

- Admin sets up academic structure and timetable
- Teacher starts class and initiates attendance
- Attendance captured via face recognition or QR scan
- Manual attendance used only if required

- Attendance stored with method and timestamp
- Students view attendance in real time
- Teachers upload marks
- Admin verifies and publishes results

11. Conclusion

This AI-Based Smart Attendance and Academic Management System provides a robust, scalable, and secure solution for modern educational institutions. By combining face recognition, QR-based attendance, and manual fallback, the system ensures high accuracy, prevents proxy attendance, and supports real-world constraints.